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Region region-0; provider order 0.

Region region-1; provider order 1.

heer, d gives \(\frac{d^2\pi xd}{2}\) for the sum of all its particles. But as this is to be multiplied by the sum of the squares of all the distances of the particles from the centre C, we shall have \(fp r^r=a^{t+x}x\lambda d\) . But the number of particles in the cylinder, or the mass \(\mu\) of the cy- linder, is \(a^\alpha\pi\lambda d\) , therefore, substituting \(\mu\), instead of this value of it in the preceding equation, we have \(fpr^r=\frac{\mu a^2}{2}\), and, dividing both sides by _n_, we have

Region region-2; provider order 2.

\(\int_{pr^r}^n}{\nu}=\frac{\mu a^2}{2n}\) and, extracting the square root and mul- tiplying by 180, it becomes

Region region-3; provider order 3.

\(\sqrt[\rho r^r]\big|\xi\times 18o^s=\frac{a^1}{2n}\big|\xi\times 18o^s\). Therefore

Region region-4; provider order 4.

\(T=\frac{\mu a^2}{2n}\big|\xi\times 18o\), and since \(\int_rp^{pr^r}=l/g\), \(\frac{i}{g}\), \(\frac{\mu a^2}{2n}=\) _l g_, and by reduction \(\frac{v-\sum\mu q}{2L}\). But _g_ as is the weight W of the cylinder, therefore, by substituting W instead of _g_ \(\mu\), we obtain _n_=P^{_a_}/2L, very simple formu

Region region-5; provider order 5.

Ia for determining the value of n from experiments.

Region region-6; provider order 6.

If it were required to find a weight Q, which, acting at the extremity of a lever I, would have no momentum equal to the momentum of the force of torsion when the angle of torsion is A–M, we must make Q U+2A09 L=nX A–M.

Region region-7; provider order 7.

346. In the preceding investigation we have suppo- soil, what is conformable to systemically, that the forces of torsion is proportional to the angle of torsion, which gives us _n_xxA–M for the momentum of that force. Let us now suppose that this momentum is altered by any quantity S, then the momentum of the force of torsion will become _n_xxA–M–S', and the general equation will assume this form

Region region-8; provider order 8.

nXA–M–S U+2573 :=u \(\int(p r^a^a)\),

Region region-9; provider order 9.

and by multiplying in place of _t_ its value \(\frac{oM}{a}\), and

Region region-10; provider order 10.

taking the flient, we have

Region region-11; provider order 11.

\(n\times 2\Lambda Mm-M^{-2}\int 8M\equiv d'\frac{d'r^a}{a^k}\)

Region region-12; provider order 12.

Now, in other to find the value of P or a complete oscillation, we must divide the oscillation into two parts, the first from B to A, where the force of torsion accelerates the velocity _u_, while the retarded force, arising from the resistance of the air and the imper- flection of elasticity, diminishes the velocity _u_, and the

Region region-13; provider order 13.

second from A to B', where the force of torsion, as well as the other forces, concerner in diminishing _n_ or retarding the motion.

Region region-14; provider order 14.

347. _Ex._ 1. If S=m×A–M', we shall have for the state of motion in the first portion BA

Region region-15; provider order 15.

\(n\times 2\Lambda M-M+\frac{2m\lambda A-M^{+1}}{+1+}\frac{zm\lambda^+1}{+1+}=\omega t\int p^a^a^a\)

Region region-16; provider order 16.

Hence, when the angle of torsion becomes equal to no- thing, or A–M=0, we have

Region region-17; provider order 17.

\(n\Delta^+\frac{2m\lambda^+1}{++1}=\UU\int p^a^a^a^a,\)

Region region-18; provider order 18.

which dividing by \(\int^p^p^a\), becomes

Region region-19; provider order 19.

\(U!=\frac{n\Delta^+\frac{2m\lambda^+1}{++1}}\)

Region region-20; provider order 20.

Let us now consider the other part of the motion from A to B', and suppose the angle AC U=M, we shall find, by calling U the velocity of the point A,

Region region-21; provider order 21.

\(nMP^a}{2}++\frac{M^kr^+r+1}=\U^{-a^*}/\sum\int^p^p^a^r.\)

Region region-22; provider order 22.

Then, by substituting instead of U its value as lately found, and taking the floants, we shall have, when the velocity vanishes, or when the oscillation is finished,

Region region-23; provider order 23.

\(\Delta-M=\frac{2m}{nx++2}\times\frac{A^{+++}M^{+1}}{+1+},\)

Region region-24; provider order 24.

and if the retarding forces are such, that at each oscilla- tion, the amplitude is a little diminished, we shall have for the approximate value of A–M

Region region-25; provider order 25.

\(\Lambda-M=\frac{2m\lambda}{n\px++i}\),

Region region-26; provider order 26.

and if the angle A–M is so small that it may be treated as a common flatened quantity, we shall then have for any number of oscillations

Region region-27; provider order 27.

\(N\times\frac{2m}{nx++1}=\frac{1}{1+}\times\frac{1}{M^{+-i}}+\frac{1}{A^{+1}}.\)

Region region-28; provider order 28.

where M represents the angle to which A becomes equal after any number of oscillations N. Hence we obtain

Region region-29; provider order 29.

\(M=\frac{1}{(N\times\frac{2m\lambda+1}{n\times+1}+\frac{1}{A^{+1}})\times\frac{1}{r+1}\),

Region region-30; provider order 30.

which determines the value of M after any number of oscillations N.

Region region-31; provider order 31.

348. _Ex._ 2. If S=m×A–M',+m×A–Mp, m' and _r_ being different values of m and _r_, we shall obtain by following the mode of investigation in the last ex- ample,

Region region-32; provider order 32.

\(n\times\Delta-M=\frac{2m}{r+1}\times\frac{A^{++}+M^{+1}}{++m}/\times\frac{A^{++}xM^{+1}}{A+M}.\)

Region region-33; provider order 33.

and if the retarding force is much less than the force of torsion, we shall have for an approximate value of

Region region-34; provider order 34.

\(n\times\Delta-M.\)

Region region-35; provider order 35.

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